

Simulation of Production Systems

MPR271

Project 13: Vaishal Ashish Solanki (2000214-5332), Aakarsh Sudhakaran (19981121-8792)

Examiner: Anders Skoogh, Siyuan Chen

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Abstract

The company Engineomatic Inc. produces steel components for exclusive engines but the recurring issues regarding the poor delivery precision and efficiency of the system have become a major concern for them. So, the company wants to rectify the issues and obtain maximum efficiency within an investment budget of \$150,000. This report addresses the above-mentioned issues by analysing the dynamics of flow in the production unit, finding weaknesses, and finally proposing possible investment strategies within the constraints. The study is carried out using Siemens Tecnomatix® Plant Simulation software, which is a discrete event simulation software that provides statistical analysis to optimize the production flow. A conceptual model of the real-world system is built in the software and analysis is fundamentally carried out by eliminating bottlenecks in each step and ensuring streamlined throughput. Proposed total 17 improvements were done in 10 stages of bottleneck analysis within the specified budget limit. While implementing these strategies it is observed that there is almost a 126% increase in productivity. Results can have variations as far as some factors are not easily applicable in the software and the software uses various assumptions, which opens a way for further studies upon the same.

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1. Introduction

1.1. Background

Engineomatic Company producing steel components for high- end engines. The company is currently experiencing several issues like “Poor delivery accuracy of finished products”. The management team hypothesizes that both outdated equipment and a lack of production planning are to blame for the frequent production interruptions. They implemented LEAN concepts through a consultant to dive into the root causes, but the output was underwhelming [1].

1.2. Purpose

The purpose of this project/case is to increase the output of the production system within the given budget. The budget for the Investment is \$150,000 [1].

1.3. Objectives

Analysing the dynamics of product flow within the factory, identifying its flaws, and outlining potential investment opportunities within the constraints of the predetermined budget [1].

1.4. Problem Formulation

1.4.1. Problem identification

Keeping in mind, the assumption of the management team i.e. The root causes of frequent production disruptions are,

1. Lack of modernized infrastructure
2. Insufficient production planning.

1.4.2. Problem selection

The management team invested in a consultant in the hope of implementing LEAN concepts but the high investment costs including the pay cheque of the LEAN consultant, Shorter-lead times, and fewer buffers make the outcomes unsatisfactory. A Production planner at Engineomatic has performed a very informal study of the production throughput during the last 10 weeks. She found out that production output can vary between 85 and 155 products per week although the factory is open for production 40 hours per week. However, she used very few samples, and she is wondering if her study is valid in the long run [1].

1.4.3. Problem Definition

The company management of Engineomatic has concluded the following ultimatums

1. Impoverished, gradually degrading Infrastructure
2. Lack of delivery precision

1.5.Delimitations

Referring to the fixed budget of \$150,000. The material handling equipment at Engineomatic is considered very efficient and does not need to be included in all products of the simulation model. Thus, if there is no buffer between stations, connectors could righteously be used to describe the relations between stations in the model. Conveyors should be used when specified and could also be used to stimulate buffers. After a complete analysis of the dynamic of the product flow, the thorough improvement of the throughput of products in the factory is the main objective, however the following investigation report will serve as the decision support for future investments [1].

2. Methodology

Improving the production efficiency became the prime target for Engineomatic Inc. A discrete event simulation (DES) study is the correct tool for this study as it is used for identifying bottlenecks and potential areas for improvements in a production process. DES involves modeling the system using set of individual and specific occurrences such as a task being completed, or a product being processed. These events are represented mathematically and through algorithms, and the simulation forecast how the system performance will get affected over time [2]. DES is carried out in a software called Siemens Tecnomatix® Plant Simulation.

The study methodology can be shortly explained in few steps as listed below [3],

- **Problem formulation:** Identifying the production process of interest and its improvement goals within a budget of \$150000.
- **Data collection:** The required data is given as a part of the problem, and which is acknowledged
- **Conceptual model:** A conceptual model for the Engineomatic Inc. production system is developed which describes the system in terms of events and their sequence.
- **Simulation model:** Based on the conceptual model a simulation model is built on the Siemens Tecnomatix® Plant Simulation using various programmed commands and input data which would be able to replicate the events and their associated timestamps.
- **Verification and Validation:** Before running the simulation, the model is verified and validated by comparing its output with the real-life outputs.
- **Analysis and interpretation of results:** Various production analysis are done by using bottleneck detection method and One factor at a time method (OFAT) and drawn conclusions based on their results.
- **Documentation and presentation of results:** The methodology and results are documented in a clear and concise manner including the assumptions and limitations of the overall study.

The basic problem here is to improve the throughput of products in Engineomatic Inc factory as much as possible within an investment budget of \$150000. The required data and ideas for conceptual model was already specified as a part of the problem. The next thing is to make a plantsim model and we made the model by adding the cutting department, turning & grinding department, polishing department and surface treatment department and finally inspection & packaging department. The factory is open for production for 40 hours per week and raw material is delivered once a workday, that is for every 8 hours of production in a batch of 10 steel pipes of 42 cm length. The warehouse has a capacity of 50 cylinders, and this was indicated using a buffer of 50 capacity [1].

Moving into the cutting department the company produces 4.0 cm, 4.6 cm, and 4.9 cm long components from the raw materials in a ratio of 20%, 35% and 45% respectively. We had two options for applying the cutting method,

- i. Randomly select a number from 1 to 100 and if the number is below 21 cut 4.0 cm pieces, if it is between 20 & 56 cut 4.6 cm products and otherwise cut 4.9 cm products.
- ii. Cut 4 pieces of 4.9 cm, 3 pieces of 4.6 cm and 2 pieces of 4.0 cm from one raw material (the numbers are the closest estimated value obtained after solving algebraic relations) (See [Appendix](#))

For the first method the delivery accuracy is poor for small runs, it will be correct if we run the factory for a very long time. The Engineomatic Company has already been suffering from poor delivery precision and for the short run or if the production is topped in between; the first method exhibits variation between obtained product and required product proportion. Also, the scrap rate is high and varies each time for this method.

Alternatively, the second method produces constant product proportion after each and every raw material, in addition to that the scrap from one raw material is 0.6 cm which is the minimum we can obtain within the given conditions. Using this method, we obtain 22.22% of 4.0 cm products, 33.33% of 4.6 cm products and 44.45% of 4.9 cm products whenever we stop production, and it comes very close to the required delivery precision. So, we decided to implement the second method and it is shown in the [Appendix](#). Deviations from given input is described in model building section [3.1](#) and completed it as shown in [Figure 2](#).

The base model output matches the informal study conducted by a production planner in the company, so then we switched to analysing the issues and improving the system. Since the material handling department is efficient in the factory it was not included in the base model. Theory of constraints (TOC) is the base for this study [4]. For analysing the system, we used two indicators; initially by observing the simulation in slow speed and second one is the resource statistics. While watching the simulation we observed where the flow is limited, and products are piling up. Using the chart tool, we obtained resource statistics for each station in the system, and it displayed the amount of working time, blocked time, and failure time. Using these indicators, the components which act as the bottlenecks are identified. These bottlenecks acted as the constraints which limits the desired result as per TOC. Along with that high failure time indicates lack of proper maintenance for the component. After identifying possible bottlenecks for the production system, we followed One Factor at a Time (OFAT) method to determine the effects of each component by fixing all other components level constant [5]. So, for getting the effect of the factor, we compared the throughput before and after OFAT approach for a component through experiment manager with 100 observations. The experiment manager provided mean value and minimum and maximum values before and after each experiment. Improvements are suggested based on the bottleneck detection and running its OFAT test to see the effect in throughput [5].

3. Project Realization

3.1. Model Building

Engineomatic Company produces 3 different dimensions of product. 45% of the raw material is used for 4.9cm, 35% and 20% for 4.6cm and 4.0cm, respectively [1]. This way the company is getting different sizes of scrap whenever the process of cutting runs and it may not meet the customer requirement if the production stops because of technical issues. To deal with such a situation we changed the cutting process to produce all 3 products from each raw material using the required percentage of 45%, 35% and 20% with sustainable scrape rate of 0.6cm [6].

Tool change in Grinding Department is needed after 5000 products which is not generating in a week study, even if the efficiency of system increases at full extent, then the maximum products can produce in a week is 450 products (Total 50 Raw material) if we able to empty the warehouse at the last working day of the week. So, tool change part is not added in the model and there is only 1 workplace at Grinding department which is for Maintenance [6].

At the time of building the model, Surface Treatment were doing setup for each product instead of doing after 28 products (four cycle). By the time we got solution of setting “After last part” instead of “Before next part” we have already done the improvements that is why there is no setup time for liquid loading in the result. But setup time is added in the submission model [6].

3.2. Experimental Design

The following experimental design of Engineomatic Company’s factory signifies

1. Increment in throughput
2. Modernized layout of the factory for desired delivery precision

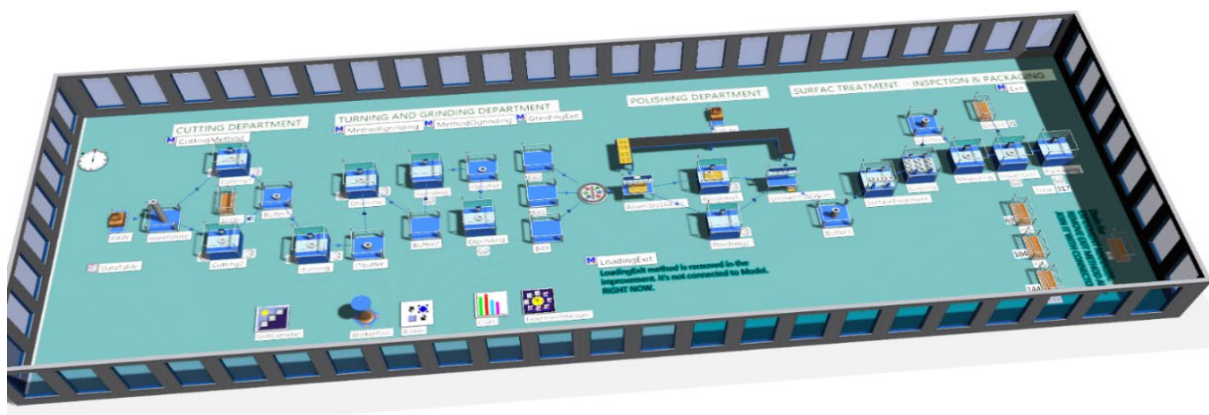


Figure 1: Improved Model (Final System, 317 products) [7].

A thorough investigation, numerous case studies, and meeting sessions with the management team resulted in the procurement of a list of appropriate modules with

reference to the budget constraints, The righteous usage of the investment budget in various aspects, and how exactly these aspects have worked out in favour of steel manufacturing giant Engineomatic Inc. has been listed as follows: -

NO.	LIST OF IMPROVEMENTS	AFTER IMPROVEMENT	BEFORE IMPROVEMENT	INVESTMENT COST (\$)
1.	Reduced cycle time at Iturning (Inner Turning) by 10%	4:39 (279 seconds)	5:10 (310 seconds)	= 9000
2.	Added Buffer (ITbuffer) after Iturning (Inner Turning)	4 (Capacity of 4)	0 (No buffer)	= 2000 (4 x 500)
3.	Reduced cycle time at Oturning (Outer Turning) by 10%	4:39 (279 seconds)	5:10 (310 seconds)	= 9000
4.	Increased buffer capacity of Buffer2	4 (Capacity of 4)	1 (Capacity of 1)	= 1500 (3 x 500)
5.	Reduced cycle time at Igrinding (Inner Grinding) by 10%	225, 252, 288 (seconds)	250, 280, 320 (seconds)	= 9000
6.	Increased MTBF at Igrinding (Inner Grinding) by 20%	168 minutes	140 minutes	= 3000
7.	Added Buffer (IGbuffer) after Igrinding (Inner Grinding)	4 (Capacity of 4)	0 (No buffer)	= 2000 (4 x 500)
8.	Reduced MTTR at Ogrinding (Outer Grinding) by 20%	13:36, 9:36, 28:00 (minutes)	17:00, 12:00, 35:00 (minutes)	= 4000
9.	Increased buffer capacity of B46	9 (Capacity of 9)	8 (Capacity of 8)	= 500
10.	Increased velocity on the fixture conveyor by 15%	$0.02505 \frac{m}{s}$	$0.0167 \frac{m}{s}$	= 6000
11.	Buy a new fixture	4 (No. of Fixture)	3 (No. of Fixture)	= 5000
12.	Increased MTBF at Polishing Stations by 50%	142.5 minutes	95 minutes	= 20,000
13.	Prepared both machines to process both dimensions	8:00, 9:30 (Uniform)	8:00 (const, Polishing 1) 9:30 (const, Polishing 2)	= 15,000
14.	Increased buffer capacity of Buffer3	8 (Capacity of 8)	7 (Capacity of 7)	= 500
15.	Added Buffer (Cbuffer) after Cooling	7 (Capacity of 7)	0 (No buffer)	= 3500 (7 x 500)
16.	Update the control system = no restarts	MTTR: 0	MTTR: 2.5 minutes	= 30,000
17.	New employee for Maintenance	2 (No. of employees)	1 (No. of employees)	= 30,000
				Total = \$150,000

Table 1. List of Investment (Chronological order) [8]

3.3. Analysis

The unsatisfactory outcome from the previously advised pathway of the consultant i.e., implementation of LEAN based concepts and the case study on throughput of Engineomatic performed by the production planner aided into the base our specified investigation [1] i.e., the areas demanding for the intensive care. The management team also conveyed their requirements in a lucid manner, so the planning made for the reformation of Engineomatic corresponding to the fixed budget was readily convenient. However, there is more than what meets the eye when we take into consideration both the theoretical aspects and the performance-based outputs in such case studies. Given below is a thorough analysis of the improvements and the promised accomplishments:

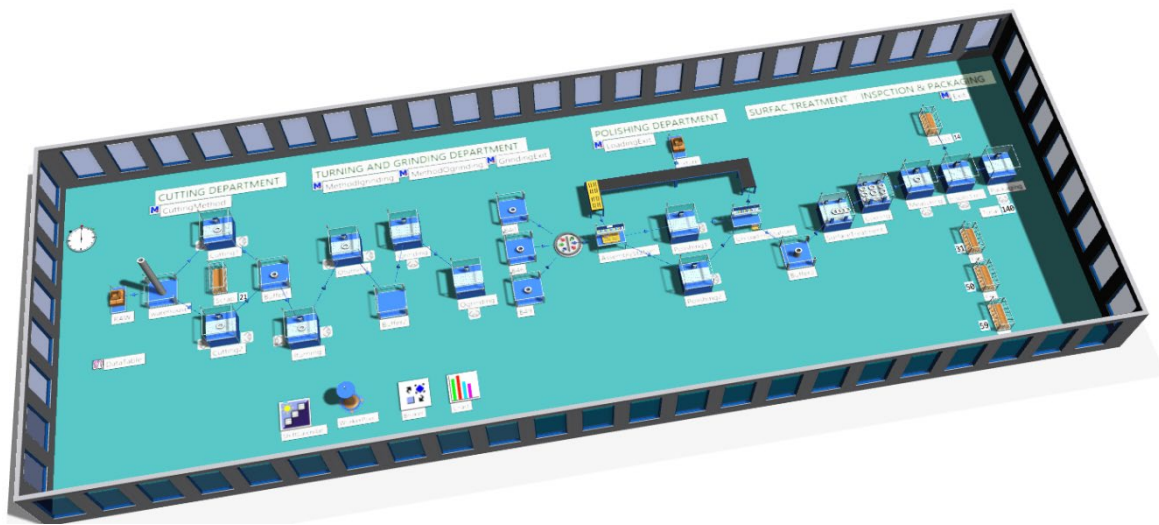


Figure 2: Base Model (140 products) [6]

From the real time analysis, it is found that Loading station is highly getting blocked because polishing machines are being unable to do work on every dimension of the products as machine 1 can handle 4.0 and 4.6 only where machine 2 can handle 4.6 and 4.9 only. Moreover, Company is producing 4.9cm in larger quantity compared to other two dimensions. So, after modernizing the machines to do every dimension of the products, a massive fall is located in blockage of Loading station as well as in the buffer after the unloading station [9].

In the factory there are a total of four operators, one for the cutting department, one for the turning and grinding department, one for quality inspection and one for the maintenance. There is only one maintenance worker in the whole factory who manages more than 1 department. Moreover, when the maintenance worker is busy at any machine and if any other machine needs maintenance, then the whole production cycle has to be stopped because there is only one maintenance worker which is busy at somewhat machine. To lessen the workload of maintenance worker, it is found that adding one new employee into maintenance department working time of all the machines are increasing and failure time is decreasing in all machines except measuring and loading, unloading and buffer after unloading station started to get blocked [9].

The reason behind this blockage may be related to the number of fixtures and speed of conveyors. As both polishing machines are modernized and able to do any dimensions of the product, new product coming to the loading station has to wait for the fixture which is on the unloading station and after unloading of the that product fixture came to the loading station. It is found more convenient to keep 4 fixtures as there are four stations in polishing department so after completing the unloading station the fourth fixture always stay on the conveyors and increasing the velocity of the conveyors it takes less time to reach loading station. After implementations of this the working time of polishing machines quite increase and the blockage vanished from the unloading station and buffer after unloading station as well working of colling and measuring also rises and blocking of cooling somewhat dropped. However, the loading station was still getting blocked but lessen compared to before [9].

Now to move more products towards the last procedure it's necessary to do an analysis of the starting phase of the production. From the real time analysis, it is found that there is only one buffer in the entire turning and grinding department which has the capacity of keeping only 1 product. This buffer is located between the turning and grinding machine so until inner grinding does not finish its work, the buffer gets blocked eventually both turning machine has to stop the production. After increasing the capacity of this buffer to carry two more products, growth has been seen in working time of turning and grinding machines as well as other production line machines. Moreover, blocking of turning and grinding machines was also reduced [9].

After doing analysis in the starting and middle phase of the production line, it has seen that the control system of Measuring machine is that old that quality inspector has to stop his inspection work and restart the measuring machine. By updating the control system failure drastically decreased at Measuring station and same with blockage at surface treatment and cooling station [9].

For Reducing the blockage of cooling station and get more products a buffer with the capacity of 7 products added right after the cooling station as surface treatment process happens in batch of 7 products and those 7 products goes to the cooling station. Because of no buffer between the cooling and measuring station all the products have to wait at the cooling station and because of that production line stops. After adding buffer, it is found that cooling station becomes blockage free moreover working time is also rise from surface treatment to packaging. Same with adding buffer after first 3 station in turning and grinding process to make production flow smooth, it is found that working time increased from turning to polishing department and blockage rapidly decrease at old buffer between turning and grinding. Moreover, somewhat reduction found in failure at turning and grinding department [9].

After getting positive results by adding buffers at turning and grinding department, all three buffers capacity is increased and made it to carry maximum 4 products and cycle time at both turning and inner grinding machine decreased by 10% but not at outer grinding as it's last station of the perspective department and after that there are three buffers with capacity of 8 products for each dimension. After implementation of this, a big difference has been found in working and failure while comparing it to the previous analysis [9].

From the simulation, it's visible to see polishing machine needs maintenance worker often. To make polishing machine work more, MTBF (Mean Time Between Failure) is increased by 50% as the higher MTBF the longer a system is likely to work before failing. After the implementation of this rapid reduction found in failure of polishing machine which resulted in blockage of loading, unloading and buffer after unloading. Moreover, somewhat rise and fall in working and failure of turning and grinding department respectively [9].

To reduce the blocking at turning and grinding department, MTBF at inner grinding increased by 20% and MTTR (Mean Time To Repair) at outer grinding reduced by 20% to reduce the time takes to resolve a failure meantime capacity of buffer for 4.6cm dimension increased by 1 product as products are taken from the grinding buffer whichever containing the highest number of the products so 35% of the raw material are use for 4.6 where 45% and 20% used for 4.9 and 4.0cm respectively in result of that capacity of buffer at unloading station also increased by 1 product to balance the system flow. After implementing this blocking at whole polishing station increased however blocking at turning and grinding department drastically decreased and working time increased throughout whole production line [9].

3.4. Analysis for TaskB

As per Engineomatic's request to verify their idea of putting individual loading and unloading stations for each polishing station [1], it is found that throughput is decreasing [10]. By investigating about its analysis, it is concluded that it is all depends on the position of workers and machines in subframe as they are coming from main model to subframe, so it is keep changing after rearranging the subframe [10]. Here is a two-experiment conducted to verify the idea of the Engineomatic in simulation: -

- First Experiment

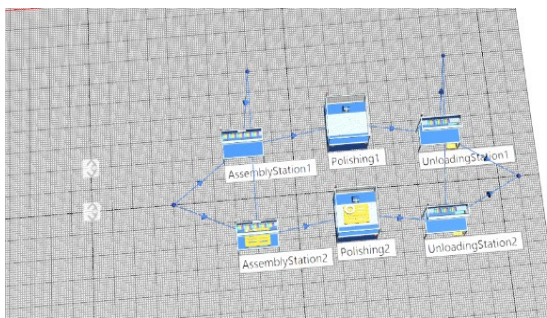


Figure 3: Subframe for first experiment [10]

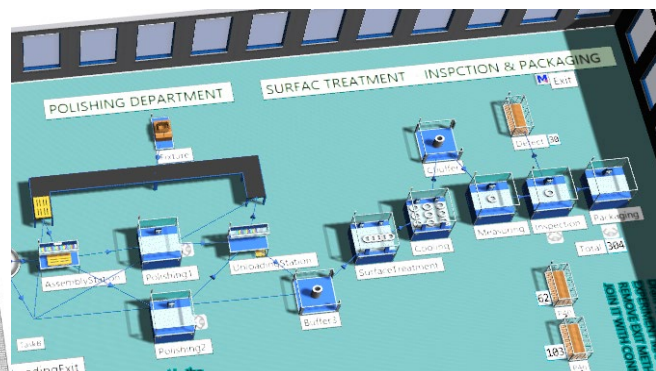


Figure 4: Main model for first experiment [10]

Total 304 products (decreased) as [Figure 1](#), 17 total products in improved model.

- Second Experiment

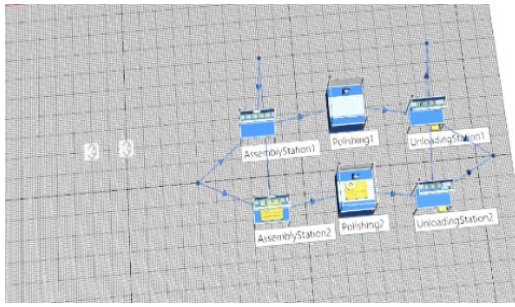


Figure 5:Subframe for second experiment [10]

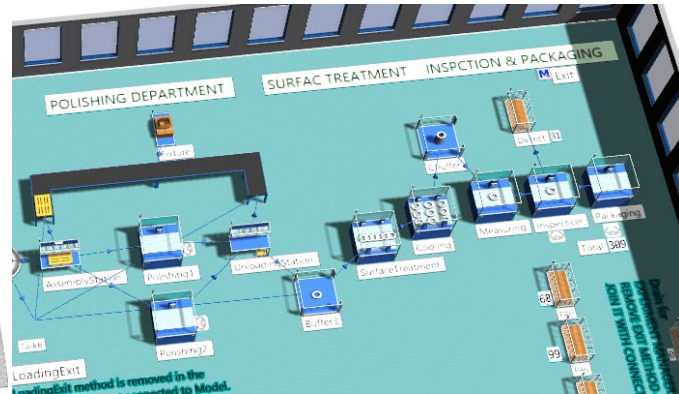


Figure 6:Main model for second experiment [10]

Total 309 products (decreased) as [Figure 1](#), 317 total products in Improved model.

It is not decreasing every time, it can be increase as well but it totally depends on where and how you place them in subframes and design of your main model as well [10].

4. Project Result

Providing the management team of Engineomatic with the most efficient infrastructure that will aid in increased throughput as well as precise product delivery within the budget constraint was the main motto behind the investigation. However, the investigation report is also to provide as a support base for any future investments made by the Engineomatic management so some crucial points like the longevity of the recently made infrastructure, and the three interconnected forms of sustainability i.e., environmental, economic, and social of the proposed renovation were also taken into consideration. Theoretically, the most competent version of output for a particular unit of input is what the study of production engineering stands for so below are the project results provided in a constructive manner: -

Base Model

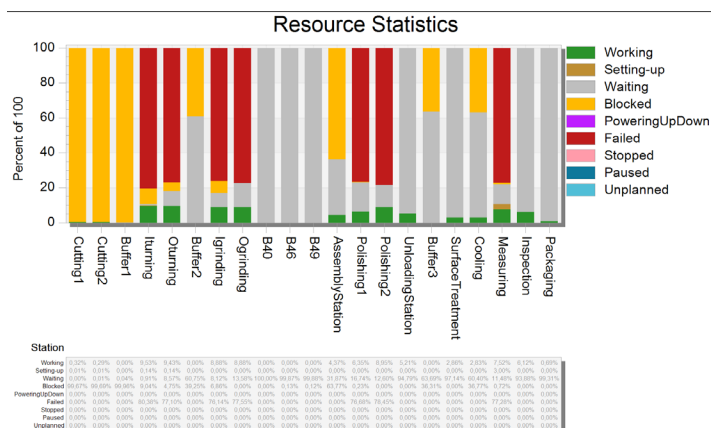
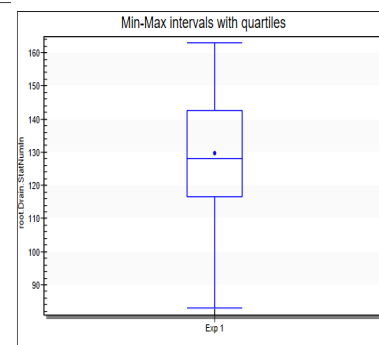


Figure 7:Resource Statistics Graph & Table (Base) [6]



Experiment	root.Drain.StatNumIn	Standard Deviation	Minimum	Maximum	Left interval bound	Right interval bound
Exp 1	129.79	17.5016853445315	83	163	126.317285930781	133.262714069219

Figure 8:Experiment Manager (Base) [6]

Improved Model

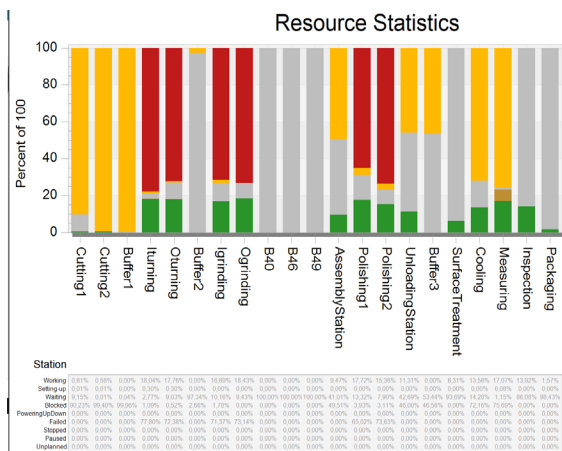


Figure 9:Resource Statistics Graph & Table (Improved) [7]

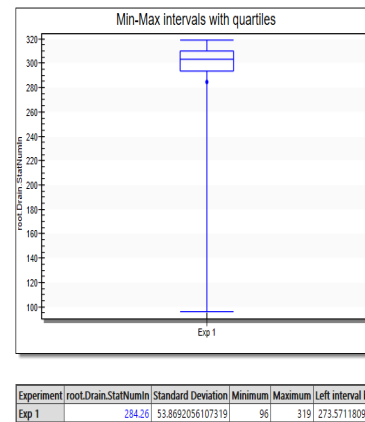


Figure 10:Experiment Manager (improved) [7]

The list of improvements resulted in increased working time of cutters from approximately 0.30% to 0.60% as well as 9.15% of waiting time is also seen at cutting machine 1 [7]. The working of turning machines increased from 9.5% to 18.04% nearly doubled and blockage is nearly become 0 after the improvements, Failure at turning machines drastically decreased from 80.38% and 77.10% to 77.80% and 72.38% [7]. Blocking of buffer between turning and grinding department went down to 2.66% from 39.25% [7]. Similar to the turning, the working of grinding is also doubled after the improvements from 8.88% to 16.69% and 18.43% [7]. Working of loading station amazingly increased from 4.37% to 9.37% and blocking decreased from 63.77% to 49.51%. The one of the major changes has seen in polishing stations [7]. The working of the station rapidly increased from 6.35% and 8.95% to 17.72% and 15.36% respectively and failure went down to 65.02% and 73.63% from 76.68% and 78.45% [7]. A massive increase in the working of the polishing department also resulted in increased working of unloading station from 5.21% to 11.31% [7]. Putting a buffer at cooling potentially increases the working of last phase of the production machines (surface inspection, cooling, Measuring, Inspection, and packaging) [3]. The biggest change in working has been observed in the cooling and measuring from the last phase of the production systems. They rose to 13.56% and 17.07% from 2.83% and 7.52% [7].

After improvements to the production systems productivity increases by 126% [6], [7]. As per Experiment manager, the mean value for the products was 129.79 and a maximum of 163 products in which the system was getting 140 products [6]. After implementing the list of improvements, the mean value went up to 284.26 and maximum 319 products in which system is getting 317 products [7].

5. Discussion

Proper production planning and methods can bring significant changes in the throughput and delivery precision for Engineomatic Inc. Some improvement methods are obvious and spotted just by looking at the productions system such as lack of maintenance employee. But most of the other improvement methods suggested in this study are not easily noticeable and it can be done through Siemens Tecnomatix® Plant Simulation software. As per this study the throughput is defined by certain constraints of the production system. Typically, bottlenecks are the factors which slows down or limits the overall output or efficiency. It can be occurred due to various reasons like limited capacity, high utilization or insufficient resource [4].

Theory of constraints (TOC) is the process methodology used in this study where bottlenecks are identified and by leveraging this constraint throughput increase is obtained. The method used to identify this constraint or bottleneck is already discussed by C Roser et al in COMPARISON OF BOTTLENECK DETECTION METHODS FOR AGV SYSTEMS [11]. In that study three different bottleneck detection methods are compared, and we make use of the method called shifting bottleneck method, which measures the active time of a component without interruption. Among the three methods discussed by C Roser et al, shifting bottleneck method shows more accuracy and reliability; but no method excels in every area as this one is hard to implement [11].

The methodology used in this work is relevant and accurate as per previous studies, and able to produce more than double of the initial throughput through various investment strategies. Also, the fact which shows how different methods can affect delivery precision is also shown in this study. However, performing a production in a virtual world has its own disadvantages such as assumptions taken, inability to incorporate all factors of real world and unpredictability of the actual condition [12]. So, the results from this study may not be on point in the case of numbers, but it can surely bring light in the right way through various further research.

6. Conclusion

With sufficient production planning and improvement methods within the budget limit of \$150,000 production efficiency improved by 126%. Poor delivery precision is also rectified by imposing proper cutting method for cutters. In addition to that, this cutting method adds value to the sustainability factors of Engineomatic as it results in constant and minimal waste rejection during production.

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Appendix

Simtalk for Cutting Method

$$4.9x + 4.6y + 4.0z = 42 \text{ where } \frac{x}{y} = \frac{45}{35} \text{ \& } \frac{y}{z} = \frac{35}{20}$$

$$4.9\left(\frac{45}{35}\right)y + 4.6y + 4.0\left(\frac{20}{35}\right)y = 42$$

$$y(6.3 + 4.6 + 2.285) = 42$$

$$\text{So, } y \cong 3, x \cong 4, z \cong 2 \text{ (4 - 4.9cm, 3 - 4.6cm, 2 - 4.0cm)}$$

```

var NewValue: real
var NewPipe: object
var len: real
len := 0.42

repeat
    waituntil Buffer1.NumMU < 3
        //Cutting 4 parts of 4.9cm from 42cm, 45%
        NewValue := len - 0.049
        if NewValue >= 0.224
            NewPipe := .UserObjects.P49.create(Buffer1)
            //decreased length of raw material after every cutting
            len := NewValue
        end
        //4 parts of 4.9cm cut from 42cm, 22.4cm remaining( 4 x 4.9 = 19.6cm) ( 42 - 19.6 = 22.4 cm)
    until len <= 0.225

repeat
    waituntil Buffer1.NumMU < 3
        //Cutting 3 parts of 4.6cm from 22.4cm, 35% ( from 42cm not from 22.4cm)
        NewValue := len - 0.046
        if NewValue >= 0.086
            NewPipe := .UserObjects.P46.create(Buffer1)
            //decreased length of raw material after every cutting
            len := NewValue
        end
        //3 parts of 4.6cm cut from 22.4cm, 8.6cm remaining( 3 x 4.6 = 13.8cm) ( 22.4 - 13.8 = 8.6cm)
    until len <= 0.087

repeat
    waituntil Buffer1.NumMU < 3
        //Cutting 2 parts of 4.0cm from 8.6cm, 20% ( from 42cm not from 8.6cm)
        NewValue := len - 0.04
        if NewValue >= 0.006
            NewPipe := .UserObjects.P40.create(Buffer1)
            //decreased length of raw material after every cutting
            len := NewValue
        end
        //2 parts of 4.0cm cut from 8.6cm, 0.6cm remaining( 2 x 4.0 = 8cm) ( 8.6 - 8 = 0.6cm)
    until len <= 0.007

repeat
    NewValue := len - 0.006
    //sending the constant scrap of 0.6cm into scrap
    NewPipe := .UserObjects.Scrap42.create(Scrap)
    len := NewValue
until len <= 0
@.delete

```

Simtalk for Inner Grinding to change Processing Time

```
//changing the processing time as per MU for Inner Grinding

if @.name = "P40"
    Igrinding.ProcTime := 250
elseif @.name = "P46"
    Igrinding.ProcTime := 280
else
    Igrinding.ProcTime := 320
end
```

Simtalk for Outer Grinding to change Processing Time

```
|
//changing the processing time as per MU for outer grinding

if @.name = "P40"
    Ogrinding.ProcTime := 250
elseif @.name = "P46"
    Ogrinding.ProcTime := 280
else
    Ogrinding.ProcTime := 320
end
```

Simtalk for Exiting the Grinding Department

```
//Sending every MU after outer grinding to it's specific buffer

if @.class =.UserObjects.P40
    @.move(B40)
elseif @.class =.UserObjects.P46
    @.move(B46)
else
    @.move(B49)
end
```

Simtalk for Loading Station to send different dimension products to specific machine

```
var Machine : object

//sending the fixture to polishing station as per which MU they are carrng
if @.cont.name = "P40"
    Machine := Polishing1

elseif @.cont.name = "P49"
    Machine := Polishing2

elseif @.cont.name = "P46"
    //checking the poishing station if they have free space because 4.6cm ring can be polish at any station
    if Polishing1.NumMU = 1
        Machine := Polishing2
    else
        Machine := Polishing1
    end
end

end

@.move(Machine)
```

Simtalk for Delivering the finished products to its specific location

```
// delivering the final MUs after packaging to it's specific storage / drain
if @.class = .UserObjects.P40
    @.move(P40)
elseif @.class = .UserObjects.P46
    @.move(P46)
else
    @.move(P49)
end
```

Simtalk for move the parts when both cutters are empty and order parts when x cylinders left in store (Task A)

1. To move the parts (ENTRY)

```
// move material when both cutters are empty

waituntil Cutting2.NumMU = 0
if Cutting1.NumMU = 0
    //check if second cutter is also empty
    if Cutting2.NumMU = 0
        @.move(Cutting1)
    else
        //wait until second cutter is empty
        waituntil Cutting2.NumMU = 0
        @.move(Cutting1)
    end
else
    // cutter 1 is not empty then wait until cutter 2 gets empty then only move
    waituntil Cutting2.NumMU = 0
    @.move
end
```

2. To order the parts (EXIT)

```
//To order material from supplier when number of cylinders left in store

if Store.NumMU < 11
    //Time taken by supplier
    Supplier.setuptime.setparam("Uniform",2:00:00,8:00:00)
    //Time taken by driver
    Scania.ProcTime.setparam("Triangle", 2:30:00, 2:15:00, 3:30:00)
    .UserObjects.P42.create(Supplier)
    .UserObjects.P42.create(Supplier)
    .UserObjects.P42.create(Supplier)
    .UserObjects.P42.create(Supplier)
    .UserObjects.P42.create(Supplier)
    .UserObjects.P42.create(Supplier)
    .UserObjects.P42.create(Supplier)
    .UserObjects.P42.create(Supplier)
    .UserObjects.P42.create(Supplier)
    .UserObjects.P42.create(Supplier)
end
```

Results:

Object	Portion	Count	Sum	Mean Value	Standard Deviation
Store	0.00%	0	0.0000	0.0000	0.0000
Cutting1	56.60%	39	3:23:04:55.6536	2:26:16.8116	3:46:34.4474
Cutting2	0.00%	0	0.0000	0.0000	0.0000

Empty Time

Object	Portion	Count	Sum	Mean Value	Standard Deviation
Store	0.00%	0	0.0000	0.0000	0.0000
Cutting1	1.03%	39	1:44:00.0000	2:40.0000	0.0000
Cutting2	1.02%	39	1:42:53.3790	2:38.2918	10.6679

Working Time

Simtalk for Rework Method (Task C)

```
var defect : integer
defect := z_uniform(1,1,100)

//3% of defective product and 2 products following that defective product
if defect < 4
  If @.name = "Reworked"
    //repeated defective products going to defect drain
    @.move(Defect)
  end
  //giving reworked name to new defective products
  @.name := "Reworked"
  @.move(Rework)
  // sending 2 following products with reworked product at rework station
  waituntil Inspection.NumMU > 0
  @.move(Rework)
  waituntil Inspection.NumMU > 0
  @.move(Rework)
elseif defect > 3
  // sending non defective products to packaging
  @.move(Packaging)
end
```

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